REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Conservation of Energy Practice Questions

Grade 12 Advanced | Module 10 Lesson 3 | Conservation of Energy

Name: _____ Class: _____ Date: _____

Answer all 30 questions. Select one option for each MCQ. Show equations, working, units and explanations for written questions.

Useful equations and guidance: Ei = Ef for an isolated system; KEi + PEi = KEf + PEf without dissipation; Emech,i = Emech,f + ΔEthermal when appropriate.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 12 book, printed pages 23–32.

P01. In an isolated system, total energy is:

A. Always decreasing B. Always purely kinetic
C. Destroyed by friction D. Conserved

P02. Mechanical energy is the sum of:

A. Kinetic and potential energy B. Mass and speed
C. Force and time D. Momentum and displacement

P03. An elastic collision conserves total:

A. Speed of every object B. Kinetic energy and momentum for an isolated collision
C. Mechanical energy only but not momentum D. Mass times height only

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P04. In a perfectly inelastic collision, the colliding objects:

A. Always rebound at equal speeds B. Have zero total momentum necessarily
C. Stick together after impact D. Retain all translational kinetic energy

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P05. When friction lowers mechanical energy, the missing mechanical energy is mainly:

A. Destroyed B. Converted into negative mass
C. Removed from every possible system D. Transferred to internal or thermal energy

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P06. A body starts from rest and descends through height 4 m on a frictionless track. Find its final speed using g = 9.8 m/s².

A. 17.71 m/s B. 4.427 m/s
C. 8.854 m/s D. -8.854 m/s

P07. A 2.0 kg block starts from rest and descends 6 m. Friction transfers 10 J of mechanical energy to thermal energy. Find its final kinetic energy; g = 9.8 m/s².

A. 215.2 J B. 53.8 J
C. -107.6 J D. 107.6 J

P08. A ball moves upward at 8 m/s. Find its maximum vertical rise using energy conservation. Neglect air resistance; g = 9.8 m/s².

A. 3.265 m B. 6.531 m
C. 1.633 m D. -3.265 m

P09. A 2 kg cart moving at +6.0 m/s collides with a stationary 4 kg cart. They stick together. Find their common velocity if external impulse is negligible.

A. 4 m/s B. 2 m/s
C. 1 m/s D. -2 m/s

P10. A 4 kg cart moving +5.0 m/s collides with a 2 kg cart moving −2.0 m/s. Afterward the first moves +1.0 m/s. Find the second cart velocity; external impulse is negligible.

A. 12 m/s B. 3 m/s
C. 6 m/s D. -6 m/s

P11. State the law of conservation of energy.

P12. When is mechanical energy conserved?

P13. Distinguish elastic from inelastic collision.

P14. Why should system boundaries be specified in an energy balance?

P15. Can a decrease in kinetic energy alone prove total energy was lost?

P16. A body starts from rest and descends through height 6 m on a frictionless track. Find its final speed using g = 9.8 m/s².

P17. A body starts from rest and descends through height 8 m on a frictionless track. Find its final speed using g = 9.8 m/s².

P18. A 2.0 kg block starts from rest and descends 9 m. Friction transfers 15 J of mechanical energy to thermal energy. Find its final kinetic energy; g = 9.8 m/s².

P19. A 2.0 kg block starts from rest and descends 12 m. Friction transfers 20 J of mechanical energy to thermal energy. Find its final kinetic energy; g = 9.8 m/s².

P20. A ball moves upward at 12 m/s. Find its maximum vertical rise using energy conservation. Neglect air resistance; g = 9.8 m/s².

P21. A ball moves upward at 16 m/s. Find its maximum vertical rise using energy conservation. Neglect air resistance; g = 9.8 m/s².

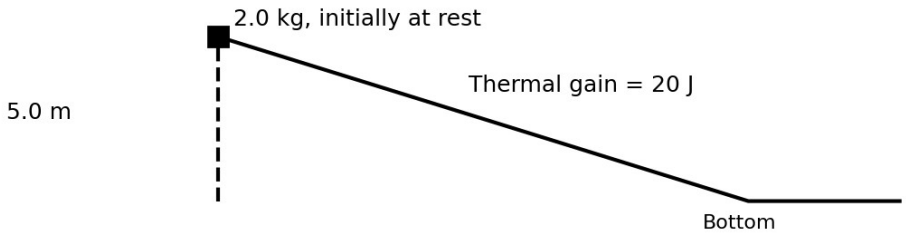
P22. A 3 kg cart moving at +6.0 m/s collides with a stationary 6 kg cart. They stick together. Find their common velocity if external impulse is negligible.

P23. A 4 kg cart moving at +6.0 m/s collides with a stationary 8 kg cart. They stick together. Find their common velocity if external impulse is negligible.

P24. A 6 kg cart moving +5.0 m/s collides with a 3 kg cart moving −2.0 m/s. Afterward the first moves +1.0 m/s. Find the second cart velocity; external impulse is negligible.

P25. A 8 kg cart moving +5.0 m/s collides with a 4 kg cart moving −2.0 m/s. Afterward the first moves +1.0 m/s. Find the second cart velocity; external impulse is negligible.

P26. A 2.0 kg block descends 5.0 m from rest and 20 J becomes thermal energy. Find final KE and speed.



P27. Two frictionless tracks have different shapes but the same starting and ending heights. An object starts from rest on each. Compare final speeds.

P28. Two equal-mass carts collide; one initially moves at 4.0 m/s and one is stationary. If they stick, find final speed and the fraction of initial KE remaining.

P29. A student uses mechanical-energy conservation for a rough ramp without a thermal term. Explain the correction.

P30. Describe an experiment to test approximate mechanical-energy conservation for a pendulum.